

From Environmental Screening to Observable Void Signatures

A Halo-Environment Interpretation of MAAT Metric Response

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Abstract

Paper 46 introduced environmental screening as an effective suppression of the MAAT projection response in dense or structurally stabilized regions. Its limitation was that the environmental variable was still mostly phenomenological. This paper turns that construction into a more observationally interpretable benchmark by rewriting the screening variable in halo-environment language: an overdensity $\Delta = \rho/\bar{\rho}$ and a bounded tidal/shear proxy $\Sigma_{\text{env}} = q^2/(q^2 + q_*^2)$.

The screened projection kernel is

$$\hat{\mathcal{C}}_{\text{env}}(z, \Delta, q) = \hat{\mathcal{C}}_{\text{proj}}(z) [1 + \alpha_{\rho} \Delta_+^n + \alpha_{\sigma} \Sigma_{\text{env}}(q)^m]^{-1}, \quad \Delta_+ = \max(\Delta - 1, 0).$$

It modulates the effective growth channel μ_{env} , the metric slip channel $\eta_{\text{slip,env}}$, and the Weyl/lensing response $\Sigma_{\text{lens,env}}$ inherited from Paper 45.

The new observable is the lensing-to-growth response ratio

$$R_{\text{LG}}(z) = \frac{\Sigma_{\text{lens,env}}(z)}{\mu_{\text{env}}(z)} = \frac{1 + \eta_{\text{slip,env}}(z)}{2},$$

which isolates the metric-slip/lensing channel from the matter-growth source. A reproducible Monte-Carlo benchmark with 6000 synthetic environments per class predicts a clear hierarchy: void-like regions are nearly unscreened ($\langle \mathcal{S}_{\text{env}} \rangle \simeq 0.999$, $\langle \max(R_{\text{LG}} - 1) \rangle \simeq 1.498\%$), cluster-like regions are strongly suppressed ($\langle \mathcal{S}_{\text{env}} \rangle \simeq 0.161$, $\langle \max(R_{\text{LG}} - 1) \rangle \simeq 0.241\%$), and a high-density local proxy returns essentially to GR ($\langle \mathcal{S}_{\text{env}} \rangle \simeq 2.13 \times 10^{-4}$, $\langle \max(R_{\text{LG}} - 1) \rangle \simeq 3.20 \times 10^{-4}\%$). The predicted void-cluster contrast is $\Delta R_{\text{LG}}^{\text{void-cluster}} \simeq 1.257$ percentage points. An explicit sensitivity check over 11 screening-ansatz variants preserves a positive void-cluster contrast in all cases, with $0.572 \lesssim \Delta R_{\text{LG}}^{\text{void-cluster}} \lesssim 1.401$ percentage points.

This is not a full halo model, a weak-lensing likelihood, or a microscopic screening derivation. Its contribution is narrower: it turns the MAAT environmental-screening benchmark into a concrete, bounded, and testable void-versus-cluster lensing/growth signature.

1 Purpose and Status

Paper 46 addressed a standard problem for any cosmological response beyond pure Λ CDM: if a metric or growth modification is visible on large scales, why does it not obviously violate local or cluster-scale constraints? The answer proposed there was effective environmental screening:

$$\hat{\mathcal{C}}_{\text{proj}}(z) \longrightarrow \hat{\mathcal{C}}_{\text{env}}(z, \Delta, \Sigma_{\text{env}}) = \hat{\mathcal{C}}_{\text{proj}}(z) \mathcal{S}_{\text{env}}(\Delta, \Sigma_{\text{env}}).$$

That construction was useful but still too internal to the MAAT formalism. The present paper addresses this weakness directly. It rewrites the environmental variable in terms of quantities

closer to large-scale-structure language: overdensity, tidal environment, lensing response, and growth response.

The status remains deliberately modest. This is a phenomenological bridge benchmark. It is not a data fit, not a Boltzmann-code implementation, not a complete halo model, and not a derivation of chameleon, Vainshtein, symmetron, or any other microscopic screening mechanism. The purpose is to identify a clean observable signature:

if MAAT metric response is environmentally screened, voids should show a larger lensing-to-growth response than clusters or local dense environments.

2 Self-Contained Projection Kernel

For readability, we spell out the projection kernel used in this benchmark rather than assuming the reader has read Papers 43–45. The kernel is a bounded diagnostic of projection stress, not a new energy density. It compares expansion-driven configuration breadth with an approximate growth-depth term.

For a flat Planck-like Λ CDM reference,

$$E(z) = \sqrt{\Omega_{m0}(1+z)^3 + \Omega_{\Lambda0}}, \quad (\Omega_{m0}, \Omega_{\Lambda0}) = (0.315, 0.685). \quad (1)$$

The expansion breadth proxy is

$$B_{\text{exp}}(z) = E(z)(1+z). \quad (2)$$

The growth-depth proxy is

$$D_{\text{acc}}(z) = D_{\Lambda\text{CDM}}(z) L(z), \quad (3)$$

where $D_{\Lambda\text{CDM}}$ is a normalized Λ CDM growth approximation and

$$L(z) = L_0 + \frac{1 - L_0}{1 + \exp[s(z - z_c)]} \quad (4)$$

is a bounded latent-depth factor. The raw projection observable is

$$\mathcal{C}_{\text{raw}}(z) = \frac{B_{\text{exp}}(z) \tanh[(B_{\text{exp}}(z)/B_*)^\alpha]}{1 + \gamma_\lambda D_{\text{acc}}(z)}. \quad (5)$$

Finally, on the benchmark interval $0 \leq z \leq 3$,

$$\hat{\mathcal{C}}_{\text{proj}}(z) = \text{clip}_{[0,1]} \left[\frac{\mathcal{C}_{\text{raw}}(z) - \mathcal{C}_{\text{raw}}(0)}{\mathcal{C}_{\text{raw}}(3) - \mathcal{C}_{\text{raw}}(0)} \right]. \quad (6)$$

Table 1: Fixed projection-kernel parameters used in this benchmark.

Parameter	Value
γ_λ	1.330507
B_*	4.509058
α	1.993673
z_c	1.1
s	3.5
L_0	0.20

These parameters are fixed from the previous projection benchmark and are not refit in the present paper. Thus Paper 47 tests the environmental response of a fixed projection kernel rather than tuning the kernel to obtain the void-cluster signature.

3 From Paper 46 to a Halo-Environment Proxy

Paper 45 defined the effective growth and metric-response channels

$$\mu(z) = 1 + \eta_g \hat{\mathcal{C}}_{\text{proj}}(z), \quad (7)$$

$$\eta_{\text{slip}}(z) = 1 + \beta_s \hat{\mathcal{C}}_{\text{proj}}(z), \quad (8)$$

$$\Sigma_{\text{lens}}(z) = \mu(z) \frac{1 + \eta_{\text{slip}}(z)}{2}. \quad (9)$$

Here μ modifies the matter-growth source term, $\eta_{\text{slip}} = \Phi/\Psi$ describes the scalar metric-potential slip, and Σ_{lens} parameterizes the Weyl/lensing response channel.

Paper 46 made these responses environment-dependent:

$$\mu_{\text{env}}(z) = 1 + \eta_g \hat{\mathcal{C}}_{\text{env}}(z), \quad (10)$$

$$\eta_{\text{slip,env}}(z) = 1 + \beta_s \hat{\mathcal{C}}_{\text{env}}(z), \quad (11)$$

$$\Sigma_{\text{lens,env}}(z) = \mu_{\text{env}}(z) \frac{1 + \eta_{\text{slip,env}}(z)}{2}. \quad (12)$$

The present paper keeps this structure unchanged but gives the environmental suppression variable a more concrete interpretation. Let

$$\Delta = \frac{\rho}{\bar{\rho}} \quad (13)$$

be the local matter overdensity relative to the cosmic mean. Let q be a dimensionless tidal/shear amplitude. Formally, one may view q^2 as proportional to the squared traceless tidal tensor,

$$q^2 \propto s_{ij} s^{ij}, \quad s_{ij} = \partial_i \partial_j \Phi_N - \frac{1}{3} \delta_{ij} \nabla^2 \Phi_N, \quad (14)$$

after suitable smoothing and normalization. The present toy benchmark does not compute s_{ij} from a survey density field. Instead, it uses q as a controlled environment proxy and maps it to a bounded stability/shear variable

$$\Sigma_{\text{env}}(q) = \frac{q^2}{q^2 + q_*^2}. \quad (15)$$

The Paper-46 screening factor becomes

$$\mathcal{S}_{\text{env}}(\Delta, q) = [1 + \alpha_\rho \Delta_+^n + \alpha_\sigma \Sigma_{\text{env}}(q)^m]^{-1}, \quad \Delta_+ = \max(\Delta - 1, 0). \quad (16)$$

This keeps the response bounded:

$$0 \leq \hat{\mathcal{C}}_{\text{proj}} \leq 1, \quad 0 \leq \mathcal{S}_{\text{env}} \leq 1, \quad 0 \leq \hat{\mathcal{C}}_{\text{env}} \leq 1.$$

The form of Eq. (16) should be understood as a minimal bounded monotone ansatz. It is chosen because it satisfies four operational requirements:

- it approaches unity in low-density, low-tidal environments;
- it decreases monotonically with overdensity and tidal/shear strength;
- it remains bounded and positive for all sampled environments;
- it has a transparent GR-recovery limit when $\mathcal{S}_{\text{env}} \rightarrow 0$.

The exponents and coefficients are not fundamental constants. They control the smoothness and relative strength of density and tidal suppression and must eventually be calibrated against simulations or data. For that reason, Section 7 includes an explicit ansatz sensitivity check.

4 Observable Lensing-to-Growth Ratio

The main observable diagnostic is

$$R_{\text{LG}}(z) = \frac{\Sigma_{\text{lens,env}}(z)}{\mu_{\text{env}}(z)} = \frac{1 + \eta_{\text{slip,env}}(z)}{2}. \quad (17)$$

This ratio is useful because it isolates the metric-slip/lensing channel from the matter-growth source channel. In the present benchmark, η_g controls the growth-source deformation, while β_s controls the metric-slip channel. The quantity R_{LG} therefore measures the part of the response that would appear as a lensing-versus-growth mismatch rather than as a simple rescaling of the growth equation.

The void-cluster prediction is summarized by

$$\Delta R_{\text{LG}}^{\text{void-cluster}} = \left\langle \max_z (R_{\text{LG}} - 1) \right\rangle_{\text{void}} - \left\langle \max_z (R_{\text{LG}} - 1) \right\rangle_{\text{cluster}}. \quad (18)$$

This is the concrete output that connects the effective MAAT screening model to future void lensing, cluster lensing, and growth-consistency analyses.

5 Monte-Carlo Environment Benchmark

We generate five synthetic environment populations using lognormal distributions for Δ and q . The values are not fitted to a survey. They are chosen as broad, interpretable archetypes: voids are underdense and weakly tidal, fields are near mean density, filaments are moderately overdense and anisotropic, clusters are dense and strongly tidal, and the local-dense class is a high-density proxy for GR recovery.

Table 2: Synthetic environment populations used in the Paper-47 benchmark.

Environment	Median Δ	Median q	Interpretation
void	0.20	0.15	underdense, weak-tidal region
field	1.00	0.35	mean-density field environment
filament	5.00	1.00	overdense anisotropic structure
cluster	100.00	2.00	dense halo/cluster-like region
local dense	10^6	3.00	high-density local-test proxy

We use the same representative response parameters as Papers 45–46:

$$\eta_g = 0.02, \quad \beta_s = 0.03,$$

and the Paper-46 screening parameters

$$\alpha_\rho = 0.15, \quad \alpha_\sigma = 1.0, \quad n = 0.75, \quad m = 2.0, \quad q_* = 1.0.$$

For each population we draw 6000 samples with fixed random seed (47), evaluate the screened response on $0 \leq z \leq 3$, and report population means and percentile bands.

6 Results

6.1 Environment Hierarchy

Table 3: Main Paper-47 environment results.

Environment	$\langle \mathcal{S}_{\text{env}} \rangle$	$\langle \max(\mu - 1) \rangle$	$\langle \max(\Sigma_{\text{lens}} - 1) \rangle$	$\langle \max(R_{\text{LG}} - 1) \rangle$
void	0.998821	1.997642%	3.525803%	1.498231%
field	0.960782	1.921564%	3.390466%	1.441173%
filament	0.593284	1.186568%	2.087195%	0.889926%
cluster	0.160656	0.321312%	0.563117%	0.240984%
local dense	0.000213	0.000427%	0.000746%	0.000320%

All sampled branches satisfy positivity of the effective response channels:

$$\mu_{\text{env}} > 0, \quad \eta_{\text{slip,env}} > 0, \quad \Sigma_{\text{lens,env}} > 0.$$

The minimum stable fraction across all five populations is 1.0 in this controlled benchmark.

6.2 Void–Cluster Contrast

The central result is the environment contrast:

$$\Delta R_{\text{LG}}^{\text{void-cluster}} \simeq 1.257 \text{ percentage points}, \quad (19)$$

$$\Delta R_{\text{LG}}^{\text{void-local}} \simeq 1.498 \text{ percentage points}, \quad (20)$$

$$\Delta \Sigma_{\text{lens}}^{\text{void-cluster}} \simeq 2.963 \text{ percentage points}. \quad (21)$$

This is the concrete prediction missing from the more qualitative Paper-46 screening benchmark. The effect is not presented as detected in current data. It is a target signature: void-like environments should retain the largest MAAT metric response, while clusters and local dense regions should recover near-GR behavior.

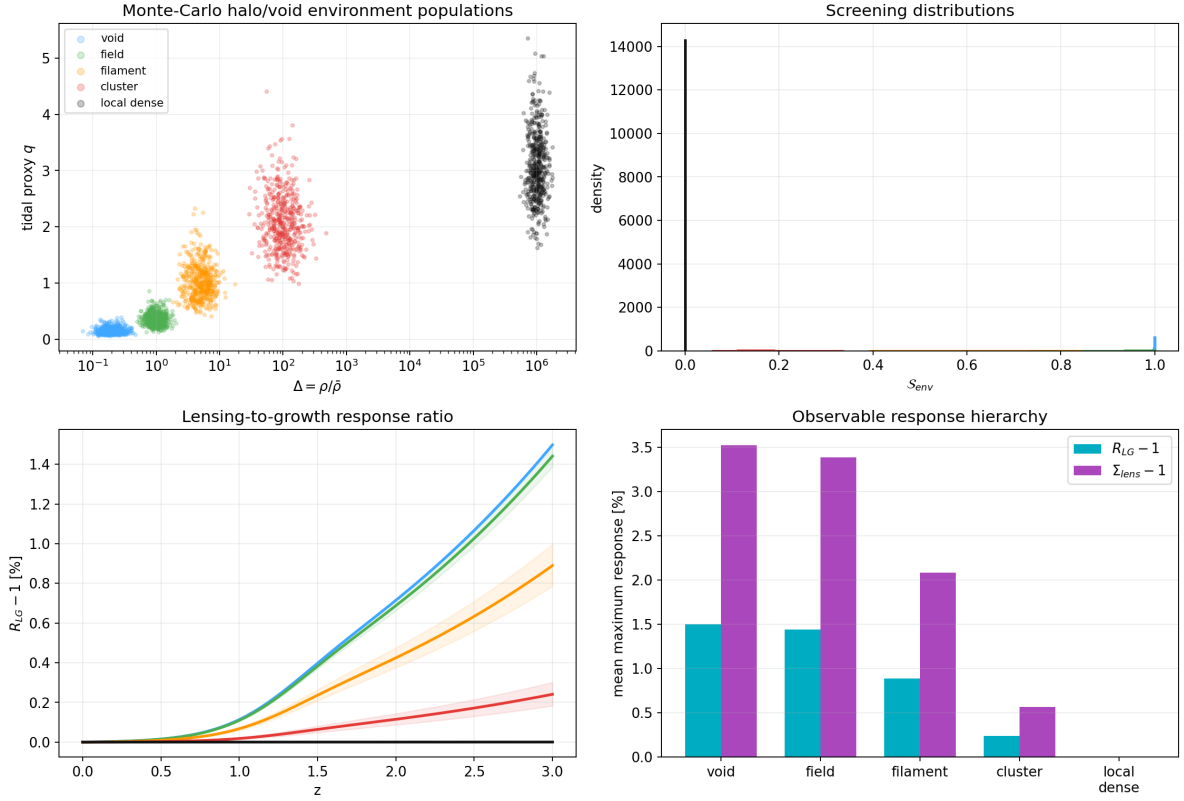


Figure 1: Paper-47 environment benchmark. Top left: Monte-Carlo environment populations in overdensity–tidal-proxy space. Top right: screening-factor distributions. Bottom left: lensing-to-growth response ratio $R_{LG} - 1$. Bottom right: response hierarchy across environments.

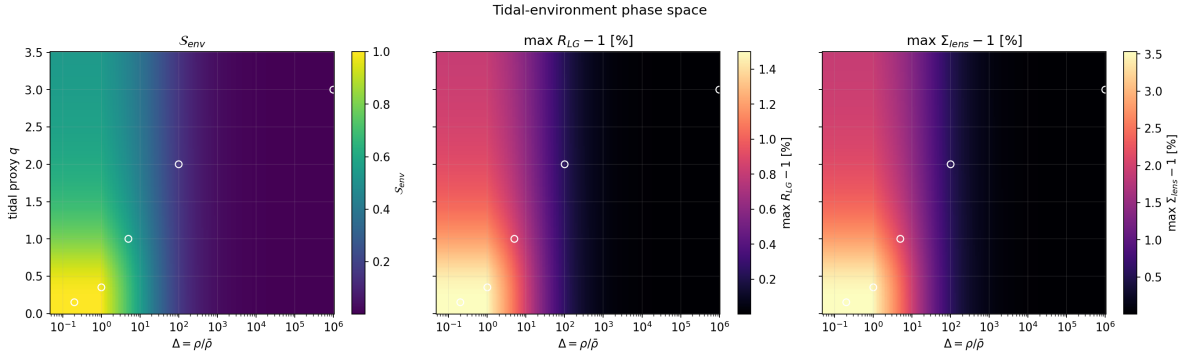


Figure 2: Tidal-environment phase space. The same environmental rule that leaves void-like regions nearly unscreened suppresses the response in high-density and high-tidal regions.

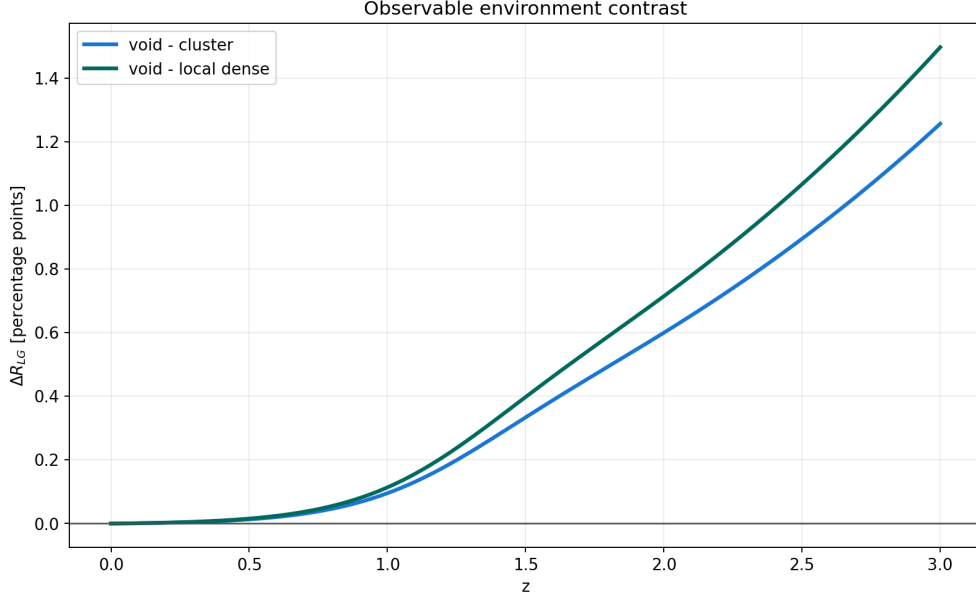


Figure 3: Observable environment contrast. The benchmark predicts a percent-level separation between void and cluster/local-dense lensing-to-growth response ratios.

7 Sensitivity to the Screening Ansatz

The screening law remains phenomenological. To quantify how much of the result depends on the chosen coefficients, we run an ablation and sensitivity check over 11 variants: weaker/stronger density suppression, softer/linear density powers, weaker/stronger tidal suppression, linear/sharp tidal powers, density-only, and tidal-only screening.

Table 4: Sensitivity of the void-cluster signature to screening-ansatz variants.

Quantity	Result
Sensitivity cases	11
Positive void-cluster contrast	11/11
Minimum $\Delta R_{LG}^{\text{void-cluster}}$	0.572 percentage points
Maximum $\Delta R_{LG}^{\text{void-cluster}}$	1.401 percentage points
Baseline $\Delta R_{LG}^{\text{void-cluster}}$	1.257 percentage points

This does not derive the screening law from first principles. It does show that the qualitative prediction is not an artifact of one specific exponent choice: void-like environments retain a larger residual lensing-to-growth response than cluster-like environments across all tested variants.

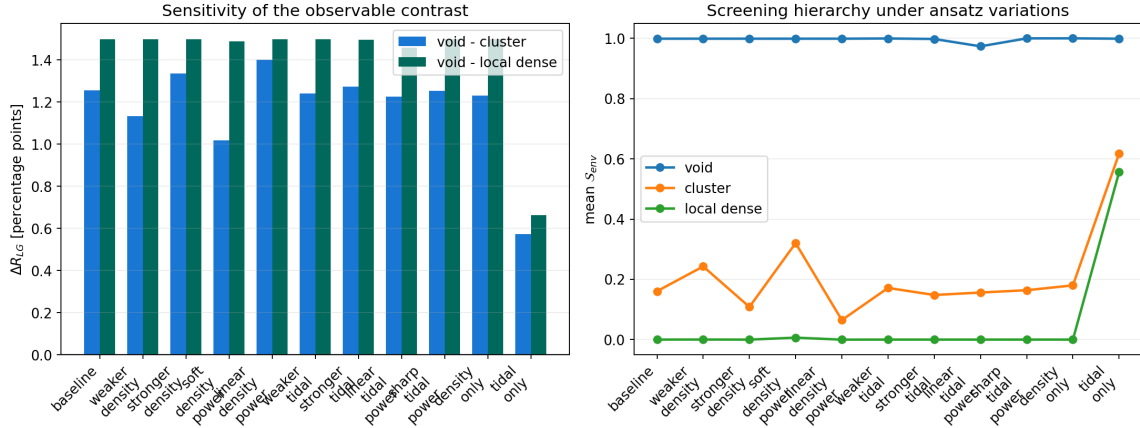


Figure 4: Screening-ansatz sensitivity. The void-cluster contrast remains positive across all tested density and tidal suppression variants, while the exact amplitude remains ansatz-dependent.

8 Interpretation

The benchmark can be summarized in one sentence:

MAAT environmental screening predicts that any residual metric response should be most visible in void-like regions and strongly suppressed in cluster-like or local dense regions.

This directly addresses the main weakness of the previous screening paper. Instead of only showing that a suppression factor can be written down, the present work identifies an observable channel:

$$R_{\text{LG}} = \frac{\text{lensing response}}{\text{growth response}}.$$

This is the type of quantity that can later be connected to void lensing, cluster lensing, redshift-space distortions, and consistency tests of gravitational slip.

The model remains phenomenological. The functions in Eqs. (15)–(16) are not derived from a microscopic action or a calibrated halo model. However, the construction is less arbitrary than Paper 46 in one important sense: the environmental axes now have recognizable large-scale-structure interpretations. Overdensity suppresses the response in dense regions; tidal/shear structure suppresses it in strongly organized environments; the remaining observable is a void-cluster lensing/growth contrast.

9 Relation to Existing Tests

Cosmic voids are a natural place to test deviations from general relativity because screening mechanisms tend to be weakest in low-density environments. Void lensing and void abundance have therefore been proposed as probes of modified gravity, while redshift-space distortions and weak lensing test the growth and metric-potential sectors more broadly.

The Paper-47 benchmark should be read in that tradition. It does not compete with full simulation-based analyses. Instead, it provides a compact effective target:

$$\Delta R_{\text{LG}}^{\text{void-cluster}} \neq 0$$

as the environment-dependent signature of a screened MAAT response.

Data-facing interpretation. Current void-lensing and modified-gravity studies already motivate comparing low-density void environments with cluster or field environments. Paper 47 does not claim agreement or tension with those measurements. It provides the quantity that should be confronted with them: the environment-dependent ratio $R_{LG} = \Sigma_{\text{lens}}/\mu$. A data-level test would require matched environment catalogues, void and cluster lensing profiles, redshift-space-distortion growth information, and their covariance. The present benchmark therefore upgrades the theory from a qualitative screening statement to a measurable target, but it is still one step short of observational inference.

10 Limitations

- The environment populations are synthetic and are not drawn from a halo catalogue or survey density field.
- The tidal proxy q is not computed from a reconstructed potential; it is a controlled phenomenological input.
- The screening law is a bounded effective ansatz, not a microscopic screening derivation.
- The sensitivity test shows qualitative robustness of the void–cluster hierarchy, but not uniqueness of the screening functional.
- No void lensing, cluster lensing, RSD, BAO, CMB, or weak-lensing likelihood is evaluated.
- The benchmark does not include nonlinear structure formation, baryonic effects, selection functions, or survey covariance.

These limitations are substantial. The result should therefore be interpreted as a falsifiable target signature for future data-level work, not as observational evidence.

11 Reproducibility

The complete experiment is included in the Structural Selection Principle repository:

Repository URL:

https://github.com/Chris4081/structural-selection-principle/tree/main/experiments/maat_void_environment_paper47

Run from the experiment folder:

```
python3 maat_void_environment_prediction_v01.py
```

The script writes:

- outputs_paper47/paper47_summary.json
- outputs_paper47/paper47_environment_summary.csv
- outputs_paper47/paper47_monte_carlo_samples.csv
- outputs_paper47/paper47_prediction_curves.csv
- outputs_paper47/paper47_tidal_phase_space.csv
- outputs_paper47/paper47_screening_sensitivity.csv
- outputs_paper47/fig1_void_lensing_growth_summary.png
- outputs_paper47/fig2_tidal_phase_space.png

- `outputs_paper47/fig3_void_cluster_contrast.png`
- `outputs_paper47/fig4_screening_sensitivity.png`

Data attribution and license note. The benchmark uses Planck-normalized reference cosmological parameters as external literature values and synthetic environment populations generated for this paper. No external survey catalogue is redistributed. All CSV and PNG files in the repository are derived reproducibility artifacts. No endorsement by the Planck Collaboration, survey collaborations, or cited authors is implied.

12 Conclusion

Paper 47 turns MAAT environmental screening from a purely internal response ansatz into an observable void-versus-cluster signature. The environmental suppression factor is rewritten in terms of overdensity and a bounded tidal/shear proxy, and the metric response is summarized by the lensing-to-growth ratio $R_{\text{LG}} = \Sigma_{\text{lens}}/\mu$.

The representative benchmark predicts that void-like regions retain an approximately 1.5% lensing-to-growth response, while cluster-like regions retain only about 0.24% and local dense regions return essentially to GR. This produces a void-cluster contrast of about 1.26 percentage points. The contrast remains positive in all 11 tested screening-ansatz variants, although the exact amplitude remains model-dependent.

The result is not observational evidence. It is a concrete, bounded, reproducible prediction target that can be tested against future void lensing, cluster lensing, RSD, and weak-lensing measurements.

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